Solution Design Document — Document Management System (DMS)

Oct 13, 2026 · School ERP, Pamantasan ng Lungsod ng Pasig · Version 0.1 (draft)

1. Document control

Version history

Version	Date	Author	Change summary
0.1	2026-10-13	Mr. Thomas Dominique G. Baran	Initial draft: scope, requirements and target architecture

Reviewers and approvers

Name	Role	Organisation	Responsibility
Mr. Joseph Wilfred Dela Cruz	Client	Pamantasan ng Lungsod ng Pasig (Client)	Reviewer: Client Requirements
Mr. Niccolo Yumul	Enterprise Architect	Malunggay Pandesal (Employee Management System)	Approver: Enterprise Solutions
Mr. Prespurt Bellen	Solutions Architect	Malunggay Pandesal (Employee Management System)	Reviewer: EMS Solution

Mr. Nathan Fernandez	Solutions Architect	Antonio Pigafetta (Payroll Management System)	Reviewer: PMS Solution
Mr. Thomas Baran	Solutions Architect	Nagpayong Pinagbuhatan (Document Management System)	Reviewer: DMS Solution
Mr. Neil Aguilar	Solutions Architect	[Team name to confirm] (Enrollment System)	Reviewer: ES Solution
Mr. Gabriel Abella	Solutions Architect	Avisala (Online Admission System)	Reviewer: OAS Solution

Related documents: Architecture diagrams (ARCHITECTURE.drawio); Business Requirements Document, DPIA, Test Strategy and Cutover Plan [to be linked].

2. Executive summary

The school ERP for Pamantasan ng Lungsod ng Pasig is made of four independent systems: Employee Management (EMS), Payroll Management (PMS), Enrollment (ES) and Online Admission (OAS). Each has its own database. There is no single, controlled place that keeps their documents and the history of their record changes.

We propose the Document Management System (DMS) as that repository. It is one Laravel 12 application with its own MySQL database and private file storage. The four systems send history to DMS through a scoped, idempotent API. Staff use a web workspace to store, share and audit documents. Every file is validated, scanned and checksummed, and every version and received change is kept. The design needs no managed cloud services, so running cost stays low. Students are not DMS users.

Decision requested: approve the target architecture (sections 7 to 10) by [date]. Hosting (section 11) stays open.

3. Business context, objectives and scope

3.1 Organisation profile: the client is Pamantasan ng Lungsod ng Pasig, Pasig City, Philippines.

3.2 Business drivers:

- Four ERP systems keep separate databases; documents and change history are not held together in one controlled place.
- Evidence must be preserved: earlier file versions, received history and audit entries cannot be overwritten.
- Integration must tolerate retries and out-of-order updates without duplicating or corrupting records.
- Running cost must stay low, so the design avoids managed database and storage services.

3.3 Objectives and success measures

ID	Objective	Measure	Target
OBJ-01	Centralized ERP document and history repository	Percentage of integrated ERP systems successfully sending documents and transaction history to the DMS	100%
OBJ-02	Safe retries and duplicate prevention	Number of duplicate history records created when an event is retried or resent	0
OBJ-03	Secure and trusted file storage	Number of files served without passing malware scanning or checksum verification	0
OBJ-04	Low operational and maintenance cost	Number of additional database engines required beyond the existing MySQL database	0

OBJ-05	Complete audit trail and accountability	Percentage of sensitive actions (uploads, downloads, modifications, deletions, and access changes) recorded in the audit log	100%

3.4 Scope

In scope	Out of scope
Document workspace with nested folders, version history, trash, restoration, and safe file previews (supported file formats to be agreed upon; TBD)	File preview support for formats outside of the agreed list
Document sharing with named recipients and comments, restricted to authorized users within the organization	External sharing, anonymous or public sharing links
Account administration and audit trail	Business approval or verification of admissions, enrollment, employment or payroll
/api/v1 boundary for history, documents, files and events	Source-side adapters and outbox in each ERP system
Storage of received transaction history and the latest snapshot of each source record	Direct management or modification of source records within ERP systems
DMS MySQL database and private version-file storage	Folder-level sharing and in-browser editing of document contents

Integration with the Employee Management System, Payroll Management System, Enrollment System, and Online Admission System for receiving documents and transaction history	Modification of business logic and databases within the ERP systems
	Students: not DMS users, no student access
	Single sign-on (to be coordinated with the Enterprise Architect)

4. Stakeholders and user personas

Volumes and peak loads are not yet known and must come from the client. Students are not users of DMS.

Persona	Approx. users	Key tasks	Primary channel	Peak load
Employee (including teaching staff)	[TBD]	Keep own files; open files shared with them; comment	Web	[TBD]
Registrar (Enrollment System, Online Admission System)	[TBD]	Store and review enrollment and admission documents; inspect received history	Web	[TBD]
HR officer (Employee Management System)	[TBD]	Store and review employee documents; inspect received history	Web	[TBD]

Payroll officer (Payroll Management System)	[TBD]	Store and review payroll documents; inspect received history	Web	[TBD]
Document Management System Administrator	[TBD]	Create accounts and roles; school repository; audit review	Web	[TBD]
ERP System Connector	4 systems	Send history and documents; read back history and events	API	[TBD]

5. Requirements

Priorities follow MoSCoW (Must, Should, Could, Won't) and are proposed for client confirmation. A full list is held in the Business Requirements Document [to be linked].

5.1 Functional requirements

ID	Module	Requirement	Priority
FR-01	Documents	Upload files as immutable versions after content validation and malware scan; keep every earlier version	Must
FR-02	Workspace	Nested folders, rename, move, star, trash and restore, safe previews, grid and list views	Must
FR-03	Sharing	Document sharing grants Viewer, Commenter or Editor permissions. Because in-browser file editing is out of scope, an Editor can rename the document and upload new versions (revision-checked). Editors cannot move, delete or reshare.	Must

FR-04	Comments	Comments on shared documents (up to 2,000 characters) and an activity sidebar	Should
FR-05	Accounts	Admin creates accounts, changes roles and activates or deactivates; no self-demotion or self-deactivation	Must
FR-06	History intake	Accept history events from EMS, PMS, ES and OAS; store an immutable receipt; update the latest snapshot only for newer revisions	Must
FR-07	History read-back	Cursor-based reads of history and events through the API; office and admin UI to inspect received versus current state	Must
FR-08	Audit	Record actor, action and document for sensitive actions	Must
FR-09	Scoped access	Limit users and connectors to their school, campus and, for office roles, their source system	Must

5.2 Non-functional requirements

ID	Category	Requirement
NFR-01	Security	Session sign-in with active-account check, CSRF protection and sign-in throttling; connector tokens stored hashed and bound to source, school and campus
NFR-02	Integrity	SHA-256 checksum recorded for every version and re-verified on access; versions are never changed

NFR-03	Idempotency	A retried request with the same key and content returns the original receipt (200). A changed request or stale revision returns 409. An older revision is stored (applied=false) without changing the latest snapshot.
NFR-04	Limits	API throttle 120 requests per minute; file up to 10 MB; history payload up to 256 KiB; folder upload up to 50 files per batch
NFR-05	Availability	[TBD]
NFR-06	Recovery	MySQL and private files restored together; RPO and RTO [to be agreed]
NFR-07	Privacy	Compliant with [Philippine Data Privacy Act of 2012 (RA 10173), to confirm]; data residency [to be agreed]
NFR-08	Auditability	All audited actions retained for [period to be agreed]
NFR-09	Accessibility	The DMS must support keyboard navigation, readable text, and accessible navigation features. Formal compliance with WCAG 2.2 Level AA is subject to client approval.

6. Current state and gap analysis

Current system	Function	Data held	Pain point / gap	Disposition

Employee Management System (EMS)	Employee management	Own database (outside DMS)	Documents and change history not held in one controlled place; source adapter not yet built	Retain and integrate
Payroll Management System (PMS)	Payroll management	Own database (outside DMS)	As above	Retain and integrate
Enrollment System (ES)	Enrollment	Own database (outside DMS)	As above	Retain and integrate
Online Admission System (OAS)	Online admission	Own database (outside DMS)	As above	Retain and integrate

Source-side adapters are planned work for each ERP team. Data-quality checks, source identity matching and per-source payload schemas have not been done yet.

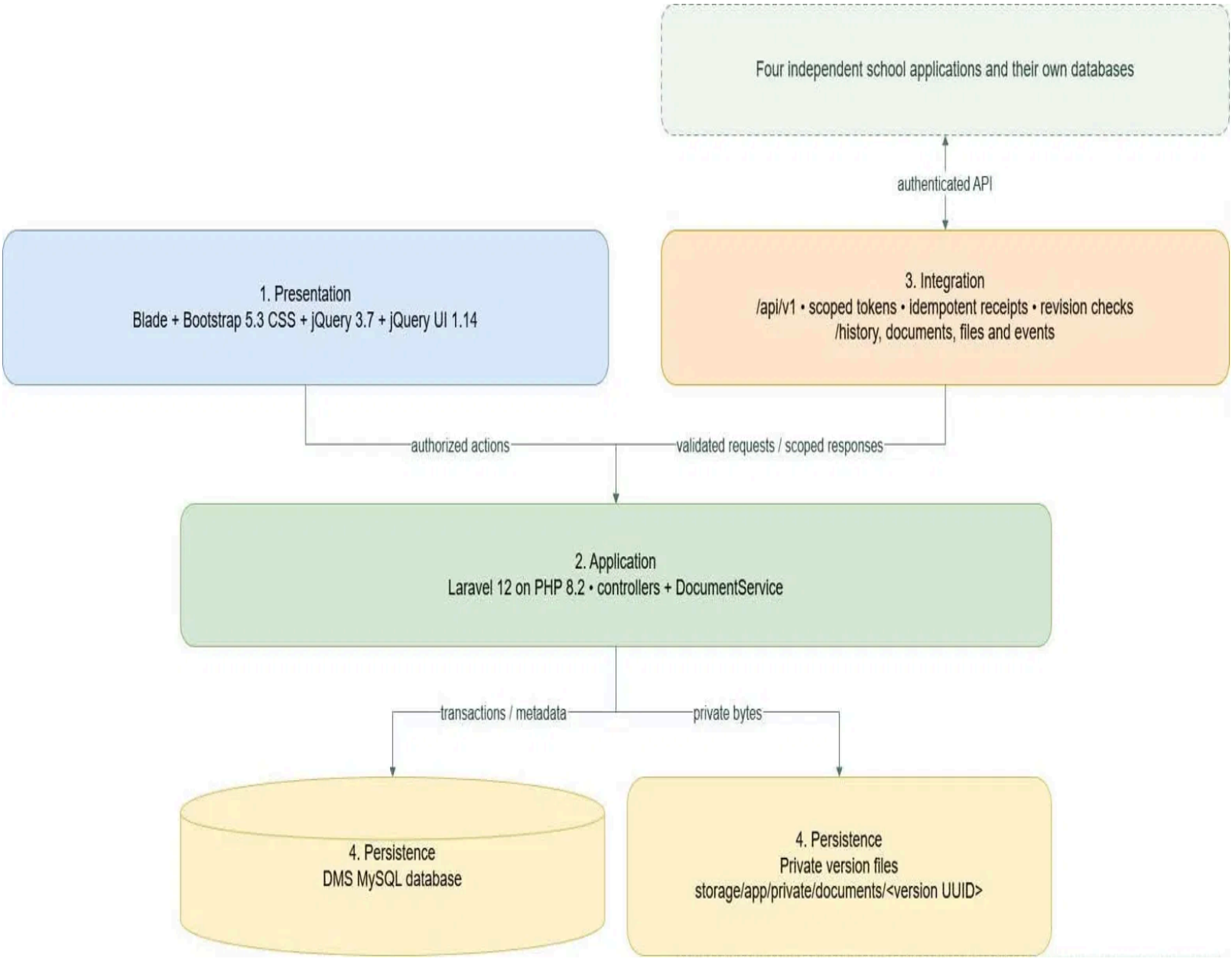
7. Target solution architecture

7.1 Architecture principles

- **Clear Data Ownership**: ERP systems manage their respective business records, while the DMS manages documents, versions, sharing, audit logs, and received history.
- **Unified Application Architecture**: The DMS uses a single Laravel application organized into four logical tiers, without requiring separate servers or microservices.
- **Controlled System Integration**: All ERP systems communicate with the DMS through the /api/v1 endpoints, without direct access to each other's databases.
- **Secure File Storage**: Files are stored in private directories outside the publicly accessible web folder, with checksums used to verify file integrity.

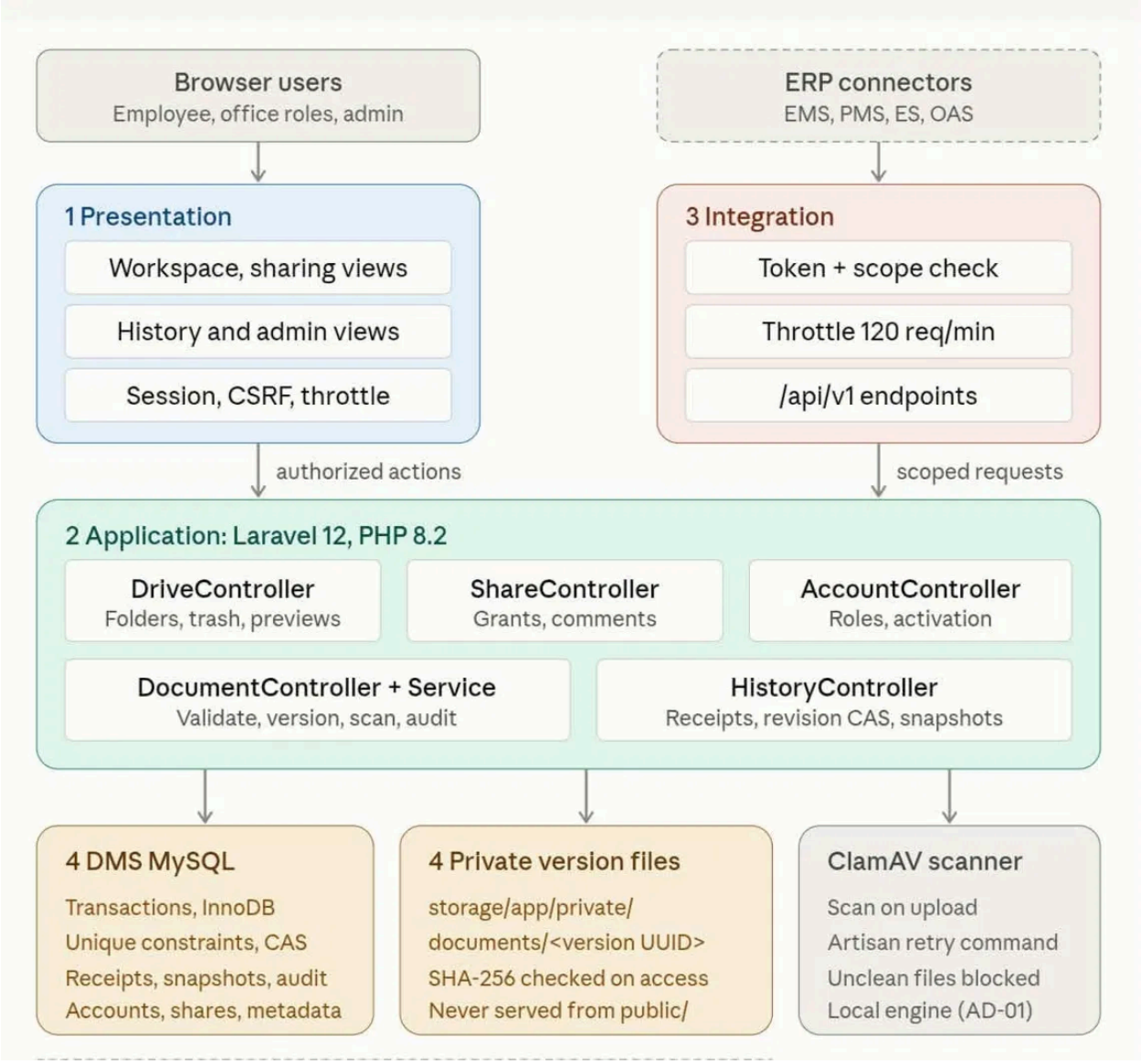
- **Reliable Data Processing**: The system prevents duplicate records and handles simultaneous requests through request receipts, revision validation, and database constraints.
- **Cost-Effective Infrastructure**: The DMS uses a single MySQL database and a straightforward folder-based storage structure to minimize operational costs and maintenance complexity.

7.2 Logical architecture



Browser users reach the application through tier 1. The four ERP systems connect only through tier 3. Tiers 1 and 3 both feed tier 2, which is the only writer to the database and the private files.

7.3 Component responsibilities



Component	Responsibility	Traces to
Presentation (Blade views, workspace)	Sign-in, role-aware navigation, folders, sharing and history screens	FR-02, FR-04, FR-07

Session security	Active account, CSRF, sign-in throttling, school/campus boundaries, grants	NFR-01, FR-09
DocumentController + DocumentService	Content validation, immutable versions, scan, idempotency, checksums, audit	FR-01, FR-08, NFR-02, NFR-03
DriveController	Folders, rename, move, star, trash and restore, previews, share creation and revocation	FR-02, FR-03
ShareController	Viewer, Commenter and Editor grants, pinned downloads, comments	FR-03, FR-04
AccountController	Admin-only account creation, roles and activation	FR-05
HistoryController	Canonical receipt hashing, revision ordering, record locks, snapshots	FR-06, FR-07, NFR-03
IntegrationAuth + API routes	Hashed token lookup, source/school/campus binding, throttle, /api/v1 endpoints	INT-01 to INT-05, NFR-01, NFR-04
DMS MySQL database	Transactions, unique constraints, revision compare-and-swap	FR-06, NFR-03
Private storage + scanner	UUID paths, SHA-256, ClamAV scan and retry command	FR-01, NFR-02, INT-06

7.4 Build vs buy: All components are custom-built on Laravel 12. Malware scanning uses a local ClamAV engine instead of a paid scanning service (AD-01).

8. Data architecture and data protection

8.1 Core entities and system of record

Entity	Key attributes	System of record	Consumers
User account	id, email, password hash, role, active, school_id, campus	DMS	Workspace
Folder	UUID, parent_id, owner, name, source, school, campus, revision	DMS	Workspace
Document	UUID, owner, folder, title, subject, reference, source, category, classification, revision, trashed_at	DMS	Workspace; ERP systems (read)
Document version	UUID, document_id, number, filename, MIME type, size, checksum, storage_key, scan status	DMS (bytes in private storage)	Workspace, shares, ERP systems
Share, comment, star	document, version, recipient, permission, expiry, revocation	DMS	Recipients
Source record snapshot	scoped record key, source, school/campus, record type and ID, revision, deleted, payload	The source ERP system (DMS keeps the latest copy)	Office/admin history UI, API read-back

Received history receipt	cursor ID, record key, event ID, revision, request hash, operation, payload, applied	DMS (immutable)	Office/admin history UI, API read-back
Audit entry, integration event	school/campus, actor, action, document, event type, revision	DMS	Admins, ERP systems (events)
Connector identity, request receipt	source, school/campus, token hash; request key, request hash, response receipt	DMS	API

8.2 Data classification (proposed; each document carries a classification field and the classes need client sign-off)

Class	Examples	Controls
Restricted	Password hashes, token hashes, payroll and HR documents	Source-scoped access, private storage, audit
Confidential	Enrollment and admission documents, received history payloads, audit entries	School/campus and source scoping, private storage, checksum
Internal	General office working files	Owner and explicit grants

8.3 Retention and privacy

- Retention schedule per entity: [to be agreed with the client, aligned to the applicable records rules].
- A Data Protection Impact Assessment is completed before go-live; reference [DPIA ID].
- Data residency: production data and backups stored in [region, depending on the hosting decision].

- ES and OAS payloads may contain personal data of students and applicants, who are not DMS users. Treat it as children's personal data and limit history access to office and admin roles.
- Payloads are source evidence, not account-provisioning instructions.
- Subject access and deletion requests: [process to be defined; not part of the current design].

9. Integration architecture

All integrations go through the one /api/v1 boundary. No education data standard is used today; the contract is a JSON format carrying event ID, record identity, operation and revision.

ID	Interface	Direction	Pattern	Frequency	Data	Standard
INT-01	EMS	EMS → DMS	REST push, idempotent	Per record change	History events, documents	Bearer token, JSON
INT-02	PMS	PMS → DMS	REST push, idempotent	Per record change	History events, documents	Bearer token, JSON
INT-03	ES	ES → DMS	REST push, idempotent	Per record change	History events, documents	Bearer token, JSON
INT-04	OAS	OAS → DMS	REST push, idempotent	Per record change	History events, documents	Bearer token, JSON
INT-05	ERP read-back	DMS → ERP (GET)	REST, cursor-based	On demand	History, events, documents, files	Bearer token, JSON

INT-06	Malware scanner	DMS → ClamAV	Command	Per upload and retry	File bytes	ClamAV command, Artisan retry

Error handling, retries and idempotency:

- New event: 201 with history_id, applied and duplicate. Identical replay: 200 with the original receipt. Same event with a different hash, or another event at the same record revision: 409.
- An older revision is stored with applied=false and the snapshot stays unchanged; a delete becomes a tombstone and earlier payloads are kept.
- Under contention DMS retries a bounded number of times, then returns 409. The source retries the same event with backoff and jitter.
- Throttle: 120 requests per minute per connector. Persistent conflicts need operator investigation; who is alerted: [to be defined].

10. Security, identity and compliance

10.1 Identity and access

- Staff sign in with school accounts (email and password hash) in a session; inactive accounts are refused.
- Admins act only within their own scope and cannot demote or deactivate themselves; new accounts inherit the administrator's scope.
- ERP systems authenticate with a bearer token stored hashed and bound to one source, school and campus.
- Single sign-on and multi-factor authentication are not in the current design [to be decided with the Enterprise Architect].

10.2 Role matrix

Role	Own files	Shared files	Source documents	Received history	Accounts and audit
Employee	Read, write	Per grant	None	None	None

Registrar, HR, Payroll	Read, write	Per grant	Own source only (ES/OAS, EMS, PMS)	Own source: inspect received vs current	None
Admin	Read, write	Per grant	School repository	Read	Manage
ERP connector	n/a	n/a	Own source, school, campus (API)	Write and read own	None

10.3 Security controls

- Transport: only the public/ directory is served, over HTTPS.
- Files: actual-content validation, 10 MB limit, private storage outside the web directory, UUID paths, SHA-256 checked on access, malware scan, and files that are not clean stay blocked.
- Application: CSRF protection, sign-in throttling, API throttle, hashed tokens, school/campus checks on every shared operation.
- Integrity: immutable versions and received history, canonical request hashes, unique constraints.
- Audit: audit entries with actor, action and document.
- Not yet decided: encryption at rest, central monitoring, penetration testing.

10.4 Compliance checklist

- Data protection law: Philippine Data Privacy Act of 2012 (RA 10173) [to confirm]
- Personal data of minors inside ES and OAS payloads assessed in the DPIA
- TLS certificates and HTTPS-only access in place
- Backup and restore of MySQL plus private files tested
- Records retention rules for the client [to confirm]
- Accessibility target [to confirm]

11. Infrastructure, deployment and operations

Status: in progress with the Enterprise Architect; not final. The hosting model is open. This section records what DMS needs whichever way it is hosted.

11.1 Hosting and environments

Environment	Purpose	Data	Hosting
Dev	Build and unit test	Synthetic	[Pending EA]
Test / SIT	Integration test with ERP adapters	Synthetic	[Pending EA]
UAT	Acceptance by reviewers and office users	Anonymised copy	[Pending EA]
Production	Live service	Real	[Pending EA]
Backup / DR	Recovery	MySQL dump and private files together	[Pending EA]

11.2 Platform (as designed)

- Compute: one PHP 8.2 / Laravel 12 application; only public/ is served.
- Data: one MySQL (InnoDB) database and a private folder for version files. No managed cloud database, queue, search or object storage is required, which keeps running cost low.
- Tenancy: school and campus scoping inside one deployment.
- Scanning: ClamAV command plus a scan-retry Artisan command.
- Delivery: CI/CD and environments [to be defined].

11.3 Operations and support

- Backups: MySQL and private files are backed up and restored together, because the database holds each file's storage_key and checksum. Restore is tested [frequency to be agreed].
- Monitoring: [to be defined]; watch files stuck in Pending, Failed or Quarantined.
- Support tiers and SLAs: [to be defined].

- The four ERP systems must be able to reach /api/v1; outages longer than their retry window lose delivery until they resend.
- Capacity: multi-writer MySQL load testing is still to be done.

12. Migration, implementation roadmap and testing

12.1 Delivery status and roadmap. Dates and phases are pending the Enterprise Architect.

Item	Status	Owner
DMS workspace, accounts, sharing, audit	Planned	DMS team
DMS /api/v1 boundary (history, documents, files, events)	Planned	DMS team
Source adapter and outbox for EMS	Planned	EMS team
Source adapter and outbox for PMS	Planned	PMS team
Source adapter and outbox for ES	Planned	ES team
Source adapter and outbox for OAS	Planned	OAS team
Full import, identity matching, payload schemas, retention, tested restore, SSO	Not started; coordinated production work	To be assigned

12.2 Data migration approach

1. Agree record types and payload schemas with each ERP team.
2. Each source commits its local change and an outbox entry together, then sends a consistent initial snapshot with increasing revisions and a stable event ID.

3. DMS stores each snapshot and receipt; the source reconciles using cursor-based reads of /api/v1/history.
4. Run rehearsal loads and reconcile record counts per source with each reviewer.
5. Cut over per source [window to be agreed with the Enterprise Architect].

12.3 Testing strategy

Test type	Scope	Owner	Exit criteria
Unit and integration	API contract: replay returns 200, altered hash returns 409, out-of-order revision stored with applied=false	DMS team	All critical paths pass
Concurrency	Multi-writer MySQL, lock and unique-constraint behaviour	DMS team	No lost or duplicated history under load
Security	Upload validation, scan blocking, token and role scoping	DMS Solutions Architect	No open critical or high findings
Backup and restore	MySQL and private files restored together	DMS team	Restored files match stored checksums
Acceptance	End-to-end scenarios per ERP system and office role	ERP Solutions Architects, office users	Sign-off by each reviewer

12.4 Training and change management: role-based guides for office staff, and an /api/v1 integration guide for the four ERP teams [to be written].

13. Risks, assumptions, decisions and appendices

13.1 Risks (ratings proposed)

ID	Risk	Likelihood	Impact	Mitigation	Owner
R-01	Hosting model not settled	Medium	High	Decide with the Enterprise Architect; keep the design hosting-neutral	Enterprise Architect
R-02	ERP adapters late, or revisions not increasing	Medium	High	Publish the API contract; DMS stores out-of-order events as applied=false; reconcile by cursor	ERP Solutions Architects
R-03	MySQL and files not backed up or restored together	Medium	High	Joint backup and a tested restore	DMS Solutions Architect
R-04	Multi-writer MySQL contention untested	Medium	Medium	Load test; bounded retries; 409 handling	DMS team
R-05	Student or applicant data inside ES/OAS payloads	Medium	Very high	DPIA; restrict history to office and admin roles	Client / DPO [TBD]
R-06	Scanner unavailable blocks uploads	Medium	Medium	Scan-retry command; monitor pending files	DMS team

13.2 Assumptions and constraints

- Each ERP team builds its source adapter and keeps revisions increasing per record.
- DMS users are school staff; students have no access.
- The client and the Enterprise Architect decide hosting, budget and data residency: [to be stated].
- Each ERP solution architect reviews the integration contract for their system.

13.3 Architecture decisions log

ID	Decision	Options considered	Rationale	Date
AD-01	Use a local ClamAV engine, run through a scan-retry Artisan command, for malware scanning instead of commercial managed scanning APIs	Third-party SaaS (VirusTotal, AWS GuardDuty); ClamAV daemon / CLI	Aligns with the low operational cost requirement (OBJ-04); works on-premise or hosted; avoids external vendor recurring costs	2026-10-13
AD-02	One Laravel deployment, custom-built, with four logical tiers	Microservices; separate API service	Small operating footprint; shared DocumentService	[date]
AD-03	One MySQL database and private folders	Managed cloud database; object storage	Lowest running cost; one backup set	[date]
AD-04	Scoped bearer tokens, idempotent receipts, revision compare-and-swap	Direct database access; shared accounts	No writes to other systems' databases; safe retries	[date]

AD-05	File scan is a security control, not business approval	Business verification inside DMS	Approval belongs to school users and originating systems	[date]
AD-06	Students are not DMS users	Student accounts and workspaces	Outside the agreed scope	[date]
AD-07	Hosting model	Local or on-premises; cloud	Open: pending the Enterprise Architect	[date]

13.4 Appendices

A. Glossary

Term	Meaning
DMS	Document Management System
EMS / PMS / ES / OAS	Employee Management, Payroll Management, Enrollment and Online Admission systems
Source	One of the four ERP systems when it sends history to DMS
Receipt	Stored result of an API request, returned again on an identical retry
Revision	Per-record number from the source that only increases
CAS	Compare-and-swap: update only if the revision is newer

DPIA	Data Protection Impact Assessment
RPO / RTO	Recovery point and recovery time objectives

B. Requirements traceability (requirement to component to test)

Requirement	Component	Test
FR-01, NFR-02	DocumentService, private storage, scanner	Unit and integration; security
FR-02, FR-03, FR-04	DriveController, ShareController	Acceptance
FR-05, FR-09, NFR-01	AccountController, session security, IntegrationAuth	Security
FR-06, FR-07, NFR-03	HistoryController, MySQL	Unit and integration; concurrency
FR-08	DocumentService, audit entries	Acceptance
NFR-06	Backup of MySQL and private files	Backup and restore